

Krzysztof Mazurek

SUMMARY

+48 735419777 | mazurekk21@gmail.com | [linkedin](#)

Enthusiastic about modeling, simulation, and control of dynamic mechanical systems, with interests in topics such as: GNC, mission analysis, multibody dynamics. I aim to deepen my expertise in numerical computing, software development and optimization.

EDUCATION

Politecnico di Milano

Sep. 2025 - Present (expected - 2027)

Master's degree in Space Engineering

Warsaw University of Technology

Oct. 2021 - Mar. 2025

Robotics and Automatic Control - Final grade : 4.4 / 5

PROFESSIONAL EXPERIENCE

Structural Analysis Engineer

May 2025 - Aug. 2025

Sener Aerospace & Defence

Warsaw, Poland

- Performed static and modal FEA of structures (ESA COMET container, ARIEL satellite components) using Siemens Femap.
- Automated FEA workflows using VBA and Python: temperature mesh mapping from Esatan to Femap, result postprocessing, midsurface preprocessing.

Software Engineer (Internship)

Jul. 2024 - Aug. 2024

Sellita Watch Company

Le-Chaux de Fonds, Switzerland

- Developed software module for Beckhoff PLC in TwinCat3- movement control of ultra-precise servo press for assembling watch components based on the user-specified criteria.
- Developed HMI for the machine control module in C#.Net framework.

STUDENT ACTIVITIES

Multibody and simulation software engineer

Nov. 2025 -

Polispace - Rover Team

Milano, Italy

- Developing the model of the rover in Simscape and its integration with Unreal Engine.

Electrical team coordinator, Mechanical team member

Jan. 2022 - Jun. 2025

Student Association for Vehicle Aerodynamics - MotoStudent Electric 2025

Warsaw, Poland

- Responsible for early-stage development of the electric drivetrain- design and validation.
- Simulated and optimized shape of a motorcycle rear pro-link suspension in Msc Adams.
- Designed a steel-tubed swingarm, chain tensioner assembly, rear brake adapter using Siemens NX.

3D printing Science Club vice-president

Oct. 2022 - Oct. 2024

Warsaw University of Technology

Warsaw, Poland

- Designed and manufactured models, customized and assembled machines (CNC, resin/ FDM printers).

OTHER PROJECTS

Spacecraft Attitude Dynamics course project

Sep. 2025 - Dec. 2025

- Attitude control of a microsat equipped with CMG, IMU, Sun & horizon sensor using LQR and nonlinear control, S/C dynamics and environment disturbances modeling - Simulink.

Orbital mechanics course project

Sep. 2025 - Dec. 2025

- Interplanetary fly-by mission design; Orbit perturbations modeling and analysis- Matlab.

Thesis - Feedforward control of underactuated multibody systems in a co-simulation framework

Oct. 2024 - Mar. 2025

- Developed a simulation environment for planar control of underactuated mechanisms using servo-constraints. It combined Python software communicating with MSC Adams via Simulink.

Satellite solar array opening simulation

Mar. 2024 - Jun. 2024

- Collaborative project with Politecnico di Milano for multibody dynamics course; using Msc Adams and Matlab.

TECHNICAL SKILLS & FOREIGN LANGUAGE

Programming: Matlab & Simulink, Python, C++

Engineering Software: Msc Adams, Siemens Femap, Ansys Mechanical & Spaceclaim, Siemens NX, Fusion 360

Tools: Git, GitHub, Linux, LaTeX, MS Office, TwinCat3

Foreign language: English C1+, German B1, Polish native